

Pictograph-to-Text Translation for Augmented and Alternative Communication

CLIN28

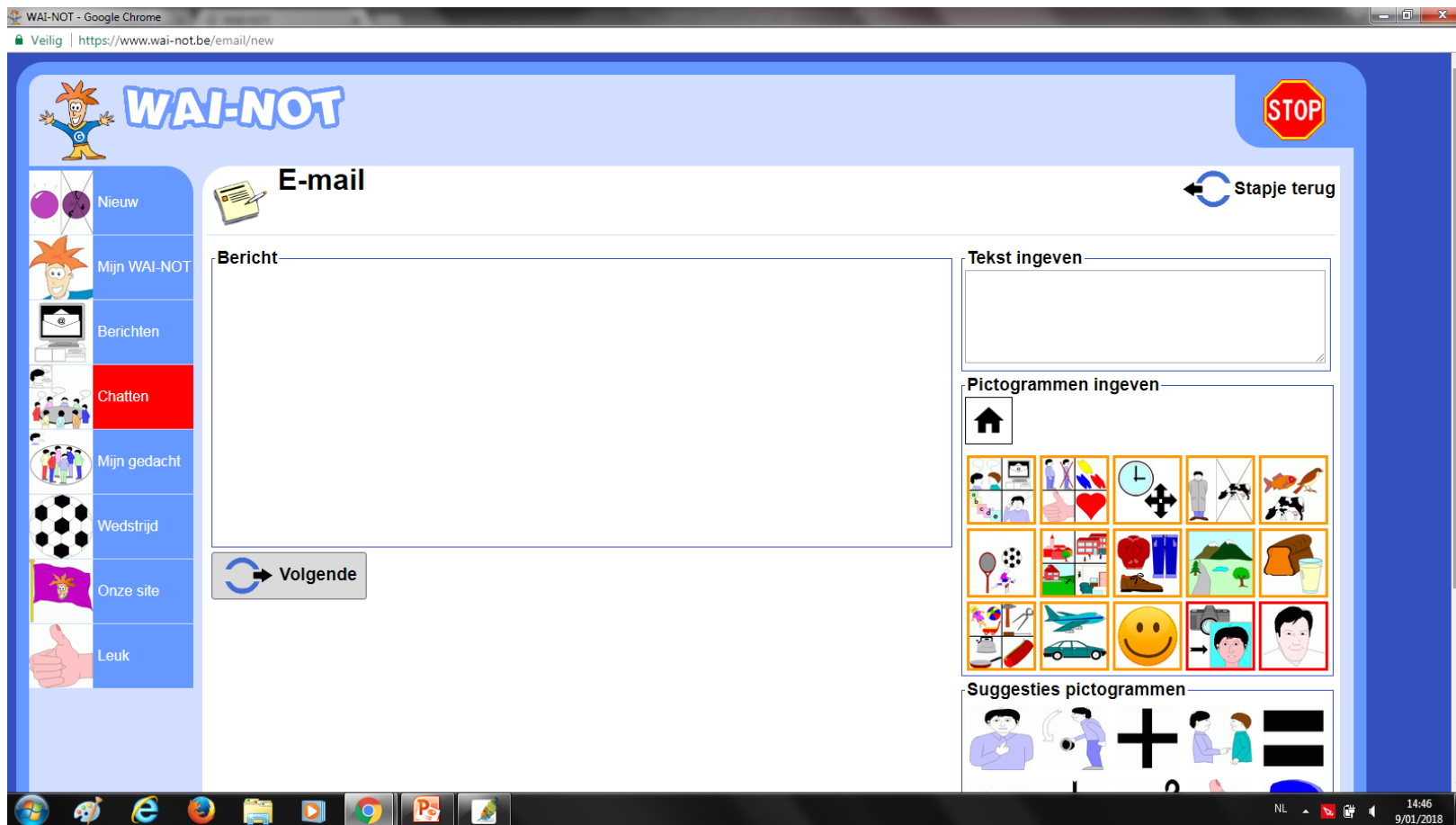
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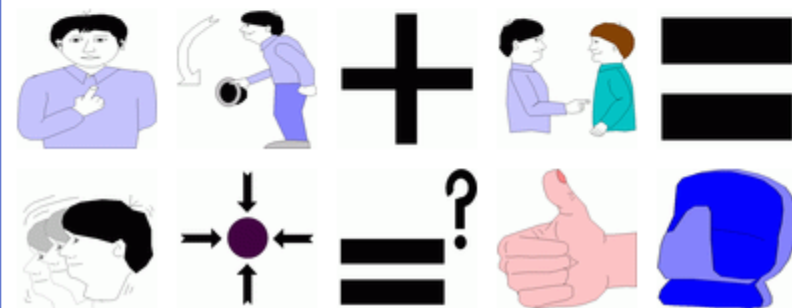
Introduction



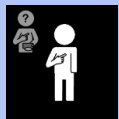
Pictogrammen ingeven



Suggesties pictogrammen



Sequence of pictographs



Sclera



Beta



Text

Mijn kat eet vis.
(*My cat eats fish.*)



The challenge...

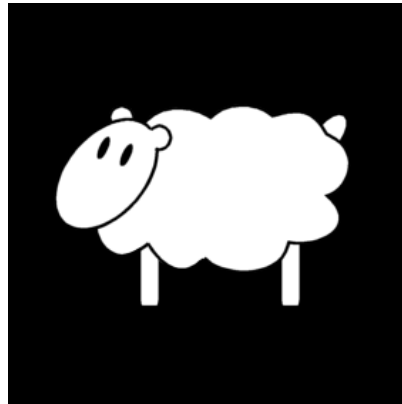
A pictograph can correspond to different (synonymous) lemmas



cash, geld, poen, cent, euro,...



Most pictographs make no distinction between singular and plural



schaap, schaapje, schapen, schaapjes,...



Verbs are not inflected, and there are no temporal auxiliaries



dans, dansen, danste, dansten, dansend,...



There are (almost) no function words (determiners, prepositions,...)

Nothing to show here



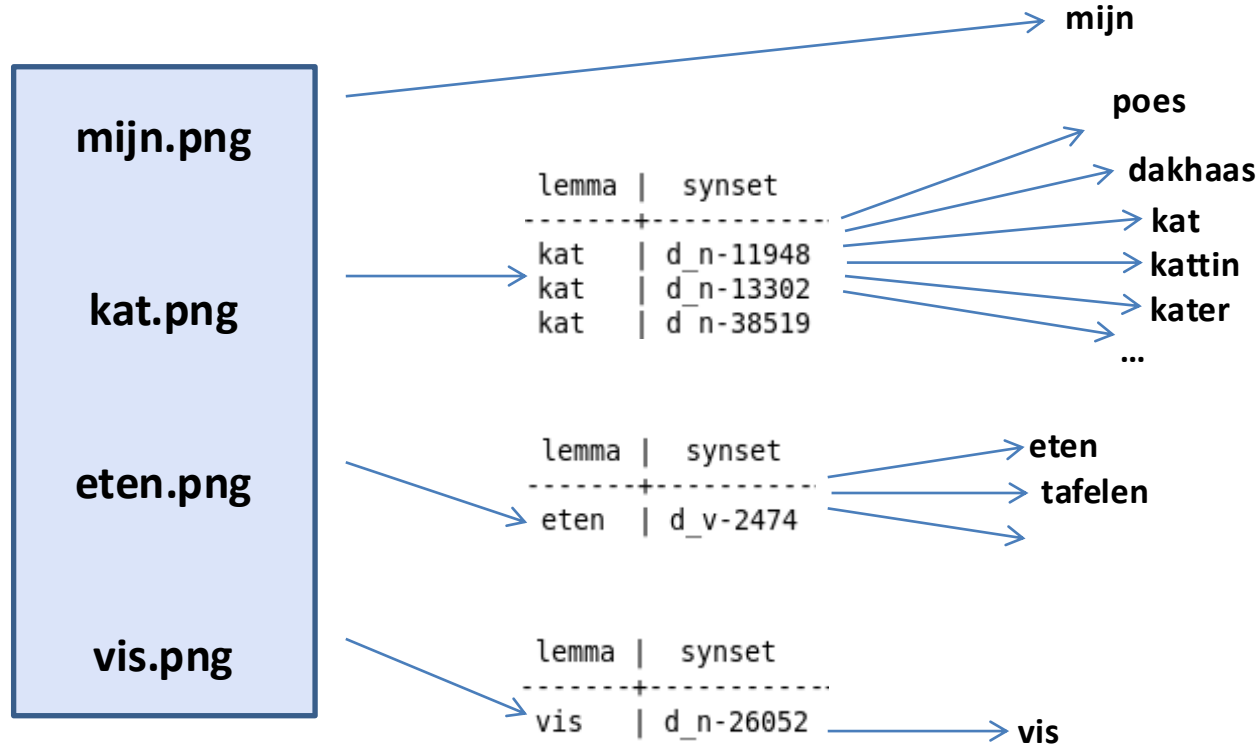
Two approaches

- **Approach #1: Language Models**
 - #1: *N*-gram-based approach
 - #2: Long Short-Term Memory-based approach (Neural Networks)
- **Approach #2: Machine Translation**
 - #1: Phrase-based Machine Translation (Moses)
 - #2: Neural Machine Translation (OpenNMT)
- (Not discussed: Rule-based systems)

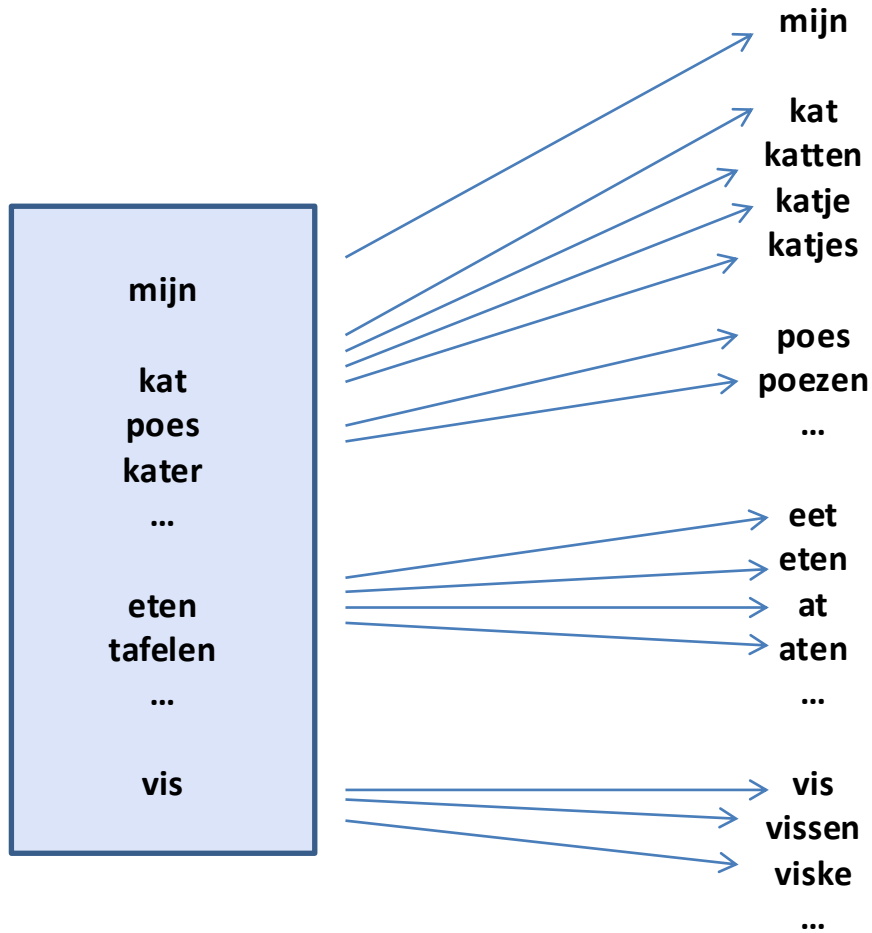


Approach #1: Language Models

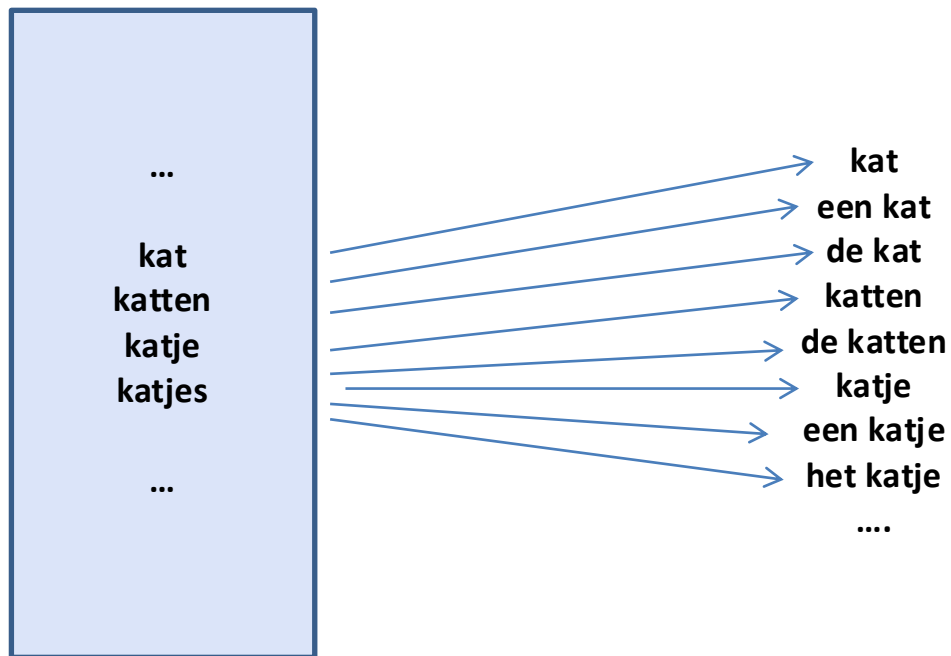
Step 1: All pictographs are connected to **synsets**. Find all synsets and retrieve all **lemmas** that are contained within these synets.

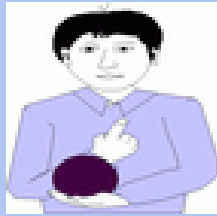


Step 2: For every lemma, generate its paradigm (**reverse lemmatisation**).

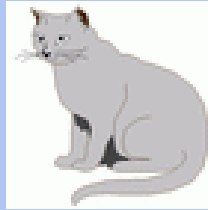


Step 3: For every noun, generate variants with/without article

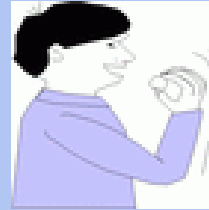




mijn.png



kat.png



eten.png



vis.png

*kat
katten
de kat
poes
de poes
....*

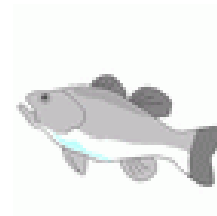
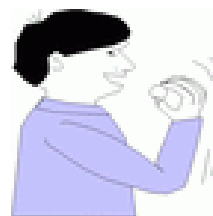
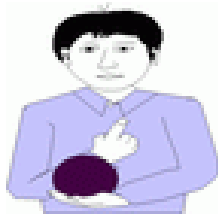
*eet
at
gegeten
tafelde
tafelen
...*

*vis
de vis
visjes
de vissen
...*



Sub-Approach #1: N-gram model-based approach

- Beam search decoding (*expands the most promising path first*)



1.	<s>				Logprob 0
2.	<s>	Mijn			Logprob -2.42
3.	<s>	mijn			Logprob -4.16
4.	<s>	Mijn	de kat		Logprob -4.01
5.	<s>	Mijn	kat		Logprob -3.52
6.	<s>	Mijn	kat	tafelt	Logprob -7.60
7.	<s>	Mijn	kat	eet	Logprob -3.47
8.	...				



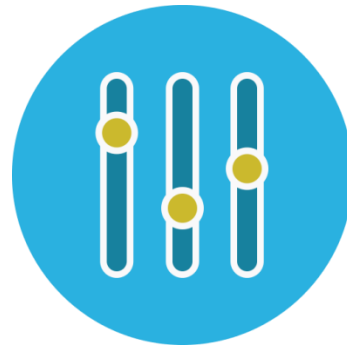
- **Model 1:** “huge” 5-gram language model
 - Interpolated Kneser-Ney smoothing
 - Trained with the KenLM toolkit on more than 1,000M tokens:

Name of resource	Reference	Number of tokens
Open Subtitles 2016	Lison & Tiedemann (2016)	266.6M
EUbookshop	Skadins et al. (2014)	245.0M
DGT	Tiedemann (2010)	72.6M
Europarl	Tiedemann (2010)	59.5M
Wikipedia	Tiedemann (2010)	19.7M
CGN (Flemish)	Oostdijk et al. (2002)	3.8M
SoNaR-500	Oostdijk et al. (2013)	500M

- **Model 2:** CGN (Flemish) (Oostdijk et al. 2002) 5-gram language model
- **Model 3:** SONAR Subtitles language model (27.6M tokens)



- Uses a number of parameters (*which are tuned with a local hill climber*):
 - **Threshold pruning:** Based on its probability, should this new path be added to the existing beam?
 - **Histogram pruning:** Sets the beam width.
 - **Cost parameter:** Estimates the cost of the pictographs that still need to be processed.
 - **Reverse lemmatiser minimum frequency:** Chooses a minimum frequency of occurrence required for a surface form to be generated.



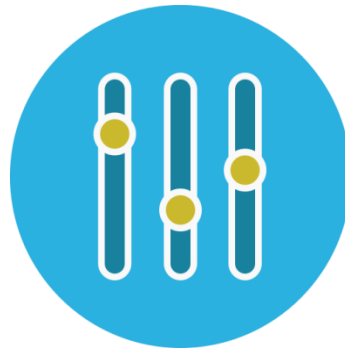
Sub-approach #2: Long Short-Term Memory-based approach

- Tensorflow (Abadi et al. 2016)
- A Recurrent Neural Network that uses a context much larger than n
- LSTM cells learn to explicitly remember or forget things from the past
- **Data:**
 - The Flemish part of CGN (Oostdijk et al. 2002)
 - (**Future work:** train model on “huge” corpus and SONAR subtitles)
 - 3.8M tokens for training, 350K tokens for validation
 - Uses the full vocabulary of the training set
 - Still high OOV rate: 19.5% (Beta) and 16.7% (Sclera)
- **Method:**
 - Generate all hypothesis translations of the full pictograph sequence
 - (Only generate most frequent lemma to avoid combinatory explosion)
 - The LSTM re-scores this hypothesis list
 - In most cases: equally highly scoring winning hypotheses



- **Parameters:**

- Two layers of 650 units each
- Trained on batches of 20 x 35 words (discourse-level)
- Network: initialised with random values and an uniform distribution between -0.05 and +0.05
- Weights: optimized with stochastic gradient descent
- The LSTM is regularised by applying 50% dropout and by clipping the norm of the gradients at 5



Sclera

	BLEU	NIST	WER	PER	Meteor	TER
5-gram Huge Corpus	3.41	2.35	63.28	58.62	25.52	65.95
5-gram CGN	4.42	2.1315	63.95	60.27	24.54	66.46
5-gram SONAR	3.73	2.10	62.40	57.66	21.53	65.02
LSTM CGN	2.94	2.27	69.80	62.75	19.96	72.57

Beta

	BLEU	NIST	WER	PER	Meteor	TER
5-gram Huge Corpus	5.53	2.51	62.60	58.53	27.03	63.59
5-gram CGN	6.98	2.85	60.17	55.52	30.56	60.72
5-gram SONAR	6.22	2.50	60.27	56	26.33	61.33
LSTM CGN	6.56	2.76	63.17	58.18	24.92	65.53



Language models: preliminary observations

- A huge 5-gram LM is not necessarily better (~~There's no data like more data~~)
 - Small domain-specific LMs outperform the huge LM
- CGN: The LSTM approach (CGN) is not necessarily better than the 5-gram approach
 - + Generating all hypotheses and rescoreing them with LSTM is slow!
- The problem with LMs in general:
 - No change in order of pictographs possible
 - We add articles to our hypotheses, but no other function words (like prepositions), so they will never show up in the output
 - Overall, the output is not fluent
- **Future work:** Human evaluations (accuracy, fluency, ranking)



Approach #2: Machine Translation

Statistical machine translation systems:

- Require a **bilingual parallel corpus** (Picto -> Dutch) to learn a translation model
- Require a **monolingual target corpus** (Dutch) to learn a language model



- **Parallel Corpus #1: Text2Picto corpus**
 - **Target:** “Subtitles” component of SoNaR (Oostijk et al. 2013) (27.6M tokens)
 - “Daily life” vocabulary (emotions, objects, etc.)
 - Includes 1st and 2nd person forms
 - Correctly written text
 - **Source:** Translate the “Subtitles” component into Beta and Sclera pictographs using Text2Picto
 - Note: possible error propagation

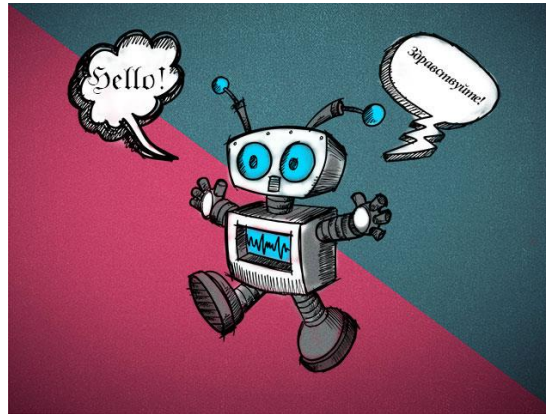


- **Parallel Corpus #2: Demolished corpus**
 - **Target:** “Subtitles” component of SoNaR (Oostijk et al. 2013)
 - **Source:** Demolish the “Subtitles” component
 - Lemmatise the corpus
 - Remove all function words and auxiliaries
 - > Make it look like pictograph input (when you replace it by its most frequent lemma)
 - Also make a factored parallel corpus which includes POS information



Sub-approach #1: Phrase-based Machine Translation:

- **Moses** (Koehn et al. 2007) (*mostly standard settings*)
- Train a model on the parallel corpus and the language model
 - Train a model for the **Text2Picto** corpus for Beta and Sclera
 - Train a model for the **demolished non-factored** corpus
 - Train a model for the **demolished factored** corpus
- Tune its parameters on 100K sentences from SoNaR Books
- Translate the test set



Sub-approach #2: Neural Machine Translation:

- **OpenNMT** (Klein et al. 2017) (*mostly standard settings*)
 - Recurrent Neural Networks
 - No linguistic knowledge (apart from *sentence* and *word*)
 - Flat translation model: uses strings only
- Train a model on the parallel corpus and the language model
 - Train a model for the **Text2Picto** corpus for Beta and Sclera
 - Train a model for the **demolished non-factored** corpus
 - Train a model for the **demolished factored** corpus
- Use 100K sentences from SoNaR Books for validation
- Translate the test set



Sclera

	BLEU	NIST	WER	PER	Meteor	TER
5-gram CGN	4.42	2.1315	63.95	60.27	24.54	66.46
Moses T2P	5.42	2.70	67.83	59.88	28.13	68.61
Moses Demolish Non-factored	5.17	2.62	76.65	64.53	23.87	75.49
Moses Demolish Factored	4.74	2.68	73.06	62.31	23.94	72.31
OpenNMT T2P	7.78	2.60	76.84	67.64	25.78	69.85
OpenNMT Demolish	5.96	2.57	78.20	67.25	22.58	76.31

Beta

	BLEU	NIST	WER	PER	Meteor	TER
5-gram CGN	6.98	2.85	60.17	55.52	30.26	60.72
Moses T2P	8.05	3.27	65.41	54.07	30.63	64.51
Moses Demolish Non-factored	5.68	2.94	71.51	60.37	26.22	71.28
Moses Demolish Factored	5.99	3.07	68.41	56.49	27.32	68.10
OpenNMT T2P	7.99	2.93	70.64	57.65	27.86	67.69
OpenNMT Demolish	7.19	2.76	74.90	64.05	25.20	77.23

Machine translation: preliminary observations

- The machine translation approaches outperform the language model approaches
 - Change in order of pictographs possible
 - MT can generate function words
 - The output is more fluent
- OpenNMT: performs equally as good as / better than Moses
 - But: “neural babble” may occur!
- The T2P corpus outperforms the demolished corpus
- **Future work:** Human evaluations (accuracy, fluency, ranking)





5-gram LM huge: Hallo je prachtige honden hoe naam hond ben Lassie?

5-gram LM CGN: Hallo je mooie hond hoe naam hond is Lassie?

5-gram LM SONAR: Hallo jij Mooi honden Hoe naam hond is lassie?

LSTM CGN: Hallo jij mooi de honden hoe de naam de honden zijn Lassie?

Moses Demolish Non-fac: Hallo zoals jij hebben een mooie hond, hoe het op naam van de hond is een Lassie?

Moses Demolish Fac: Hallo jij hebben een mooie hond, hoe in de naam van de hond is Lassie?

Moses T2P: Hallo, een mooie hond, hoe heet het hondje is een Lassie?

OpenNMT Demolish: Hallo, jij hebben mooie hond, hoe de naam van de hond is, Lassie?

OpenNMT T2P: Hallo, je hebt een mooi hondje hoe heet de hond is dat een Lassie?





Thank you!

Got any cool ideas?

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Check out our publications!

<http://picto.ccl.kuleuven.be>

